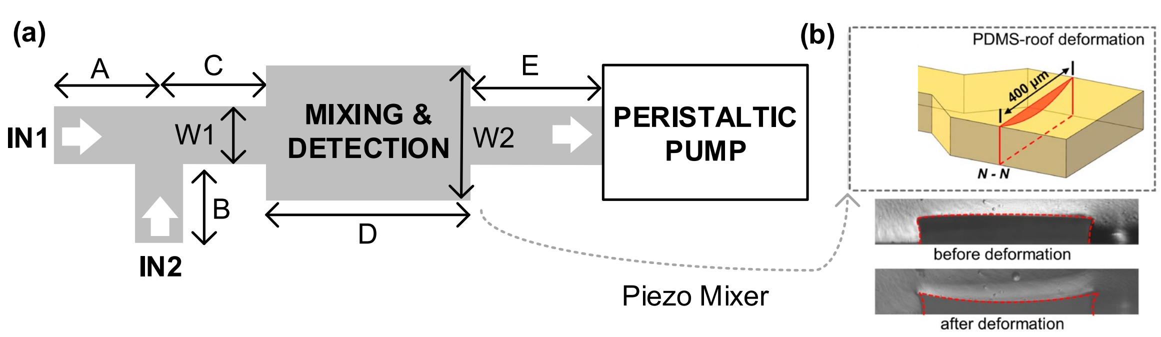
Politecnico di Milano
Biochip A.Y. 2022/2023 - M. Carminati
Exam of 25.01.2024

Test duration: 2 hours – No support material (textbooks, laptops, smartphones, notes) admitted.
All the answers must be motivated. Beyond the final numerical results, all the expressions must be provided.

Exercise 1

Consider the following microfluidic device with two inlets for analysis of DNA in liquid samples Fig.(a). The device is made of PDMS and operated in aspiration by means of an on-chip *peristaltic* pump. Mixing and detection are performed in a single chamber whose roof is flexible Fig.(b) and actuated by a piezoelectric actuator for mixing. The height (h) is constant everywhere.

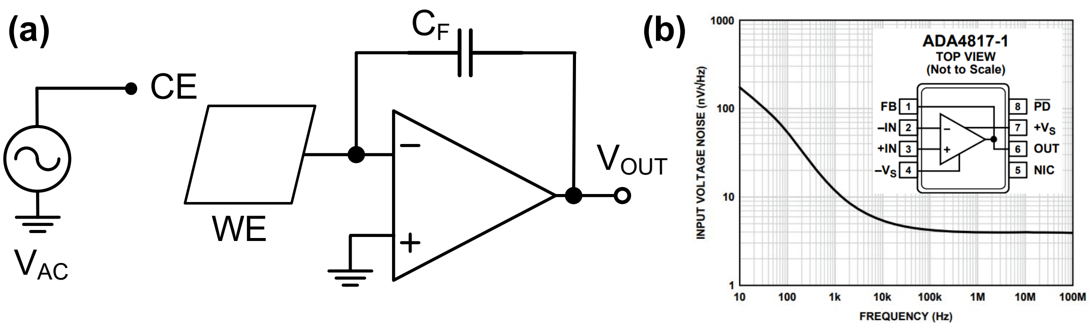


h = 100 µm
$W1$ = 100 µm
$W2$ = 400 µm
A = 1 mm
B = 1 mm
C = 2 mm
D = 5 mm
E = 2 mm
1 atm = 101 kPa
$\eta_{water} = 10^{-3}$ Pa·s

a) Draw and compute the electrical equivalent of the fluidic circuit (including the pump) assuming it filled with water. What it the impact of the deformable roof?
b) Assuming inlets at 1 atm and the pump settings flow rate of 1 µL/min, find the water speed in all sections and the value of the pressure at the input of the pump (end of section e).
c) Explain the working principle and a possible implementation structure of an on-chip peristaltic pump and briefly discuss its pros and cons.
d) Describe the working principle of a piezoelectric actuator.
e) Describe possible solutions (pros vs. cons) for mixing fluids on chip.
f) Describe briefly the steps of the fabrication process of the device (PDMS on glass).

Exercise 2

Now focus on the sensing part of the system. Consider a working electrode fabricated in gold on the bottom glass of the detection chamber used for *capacitive* detection of DNA. A capacitive feedback is adopted.



V_{DD} = +5 V
V_{SS} = -5 V
$\varepsilon_0 = 8.86\ 10^{-12}$F/m
C_{DL0} = 0.1pF/µm^2

a) Explain the working principle of a capacitive DNA sensor (assume a very large counter electrode to make its impedance negligible).
b) Assuming DNA in PBS solution, find the maximum area of the working electrode that is compatible with a max. input capacitance of 50 pF.
c) Neglecting the solution resistance and assuming 50 pF of max. capacitance, size V_{AC} and C_F for a correct detection.
d) What key element is missing for the correct operation of the preamplifier?
e) Considering the opamp voltage input equivalent noise shown in Fig.(b), study the noise of the designed circuit. Is there a preferable operating frequency to measure the capacitance?
f) Draw an alternative *ratiometric* configuration for the measurement of capacitance and compare it with the circuit used here.

Multichance students with 30% reduction of the exercises please skip points e) and f) of both exercises.